



Deltas

Beyond the coast, any mud and sand carried into the sea by the river water is usually swirled away by waves and currents. But if the load of sediment carried by the river is very large, or the sea is very calm, the sediment settles before it can be carried away. It builds up in layers that grow outward from the shore, creating an extension of the land called a delta.

SPREADING FANS

A typical river delta is a huge flat area of sand and mud. Sediment dropped by the river blocks its original course, forcing the river to spill out in many smaller channels. These soon get blocked up too, so the mud-loaded water overflows and forms even more channels, fanning out over a growing mass of soft sediment.



▲ GANGES DELTA

This satellite image shows the Ganges Delta in eastern India and Bangladesh. Every year, the Ganges River carries about 2 billion tons of mud. Much of this is deposited into the sea, where it forms a submarine fan over the Bay of Bengal.

One of the many channels flowing off the Mississippi River Delta toward the sea

River water breaking through the levees fans out in a pattern that looks like the toes of a bird

▲ MISSISSIPPI LEVEES

Water pouring into the Gulf of Mexico from the Mississippi River is separated from the sea by levees built up from the sediment carried by the river.

BIRD'S FOOT DELTA

Some rivers, such as the Mississippi in the southern US, drop sand, mud, and other sediments along their edges. This forms raised banks called levees that grow out to sea. If the river breaks through one of the levees, part of it then flows off in another direction. This creates a delta in the shape of a bird's foot. Many of the levees forming the Mississippi River Delta were swept away by the storm surge of Hurricane Katrina that flooded nearby New Orleans in 2005.